assignment _kine1D Free-fall

This is a preview of the published version of the quiz

Started: Apr 29 at 7:52pm

Quiz Instructions

2 attempts. There is a tutorial in the dropbox

Question 1 12.5 pts

A stone is thrown with an initial upward velocity of 7.0 m/s and experiences negligible air resistance. If we take upward as the positive direction, what is the velocity of the stone after 0.50 s?

- -2.1 m/s
- -4.9 m/s
- 2.1 m/s
- 0.00 m/s
- 4.9 m/s

Question 2 12.5 pts

A ball is thrown straight up with a speed of 36 m/s. How long does it take to return to its starting point, assuming negligible air resistance?

- 15 s
- 11 s
- 7.3 s
- 3.7 s

Question 3 12.5 pts

A laser is thrown upward with a speed of 12 m/s on the surface of planet X where the acceleration due to gravity is 1.5 m/s^2 and there is no atmosphere. What is the maximum height reached by the laser?

- 18 m
- 144 m
- 8.0 m
- 48 m

Question 4 12.5 pts

A bullet shot straight up returns to its starting point in 10 s. What is the initial speed of the bullet, assuming negligible air resistance?

- 49 m/s
- 9.8 m/s
- 98 m/s
- 25 m/s

Question 5 12.5 pts

A ball is projected upward at time *t* = 0 s, from a point on a flat roof 10 m above the ground. The ball rises and then falls with insignificant air resistance, missing the roof, and strikes the ground. The initial velocity of the ball is 58.5m/s Consider all quantities as positive in the upward direction. At time t=5.97s the vertical velocity of the ball is closest to

- +175 m/s.
- -12 m/s.
- -175 m/s.
- +12 m/s.
- 0 m/s.

Question 6 12.5 pts

A ball is thrown downward from the top of a building with an initial speed of 25 m/s. It strikes the ground after 2.0 s. How high is the building, assuming negligible air resistance?

○ 50 m

○ 20 m

○ 70 m

○ 30 m

Question 7 12.5 pts

A laser is thrown upward with a speed of 12 m/s on the surface of planet X where the acceleration due to gravity is 1.5 m/s^2 and there is no atmosphere. How long does it take for the laser to reach the maximum height?

○ 11 s

○ 16 s

○ 8.0 s

○ 14 s

Question 8 12.5 pts

Human reaction time is usually greater than 0.10 s. If your friend holds a ruler between your fingers and releases it without warning, how far can you expect the ruler to fall before you catch it, assuming negligible air resistance?

○ at least 9.8 cm

○ at least 4.9 cm

○ at least 6.8 cm

○ at least 3.0 cm

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